

Respiration in psychophysiology: methods and applications

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With the use of advanced equipment, respiratory measures can unobtrusively and reliably be assessed in a variety of psychophysiological research settings. New computerized analysis techniques can break down respiration into a number of components that provide valid estimates of variations in respiratory control mechanisms in the brain stem. Thus analysed, respiratory responses may vary in at least two dimensions: (A) with regard to drive and timing aspects, and (B) with regard to the metabolic appropriateness of the respiratory response. Assessment of respiratory responses may be relevant for a broad variety of research areas, including studies of the physiological effects of mental load and stress, investigations of physiological correlates of emotions and affect, and research linking physiological responses to subjective distress and psychosomatic disorders.

Keywords: Respiration, hyperventilation, stress, methodology, respiratory measurement, respiratory analysis.

Introduction

Measurement of respiration is still rather uncommon in psychophysiological research. At first sight, this seems difficult to explain. There is an abundance of evidence closely linking respiration to a variety of phenomena, such as mental effort, emotions, personality factors, and subjective distress (e.g. Bass & Gardner, 1985a; Grossman, 1983; Grossman & Wientjes, 1989), that would seem to make this response system eminently suited for research purposes. The main reasons for the infrequent employment of respiratory measures are probably not primarily of a theoretical nature, but appear to be based on practical and methodological considerations. Traditional respiratory measurement techniques are often too intrusive and too complicated to be employed in common psychophysiological research settings, or, alternatively, if noninvasive techniques are employed, too imprecise to provide adequate assessment of respiratory responses. Another reason for the apparent lack of

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enthusiasm about respiratory measures might be that the respiratory response is not a unitary phenomenon, but varies, as we will see, in a complex manner along time and volume dimensions. In addition, those investigators who are enterprising enough to employ respiratory measures, may often be faced with speech or movement artefacts that have to be dealt with. Thus, inclusion of respiratory responses in psychophysiological studies requires measurement techniques that provide sufficient precision with regard to respiratory time and volume parameters, relatively easy applicability, and unobtrusiveness, analytic approaches that permit assessment in terms of relevant response dimensions, as well as provisions to manage artefact problems. As we will argue, such techniques have recently become available.

In this paper, we shall review some elementary aspects of respiratory physiology, discuss the respiratory parameters that can conveniently be assessed in psychophysiological studies, and examine recent developments in measurement and analysis techniques. To conclude, we shall outline the potential relevance of respiratory measures for different psychophysiological research areas by proposing, on the basis of some important recent examples from the literature, a preliminary framework for the analysis and interpretation of relationships between respiration and behaviour.

Respiratory physiology

Function of respiration

The main function of respiration is to provide adequate exchange of oxygen (O_2) and carbon dioxide (CO_2) between the organism and the atmosphere. This is accomplished by several processes: (A) rhythmic refreshment of air in the lungs (ventilation), (B) diffusion of gases through the alveolar membrane in the lungs, (C) transport of gases by the blood from the lungs to the tissues (and vice versa), and (D) exchange of gas between the blood and the tissues. These processes interact with each other and with cardiovascular processes in order to meet the highly variable metabolic demands of the organism and in order to maintain homeostasis.

Control of respiration

The organism adapts to varying external or internal demands by altering the refreshment rate of the air in the alveoli of the lungs, (i.e. by changing the rate and/or depth of breathing). Rate and depth of breathing are, in turn, controlled by a number of closely interacting neural pools in the brain stem, that together are often referred to as “the respiratory centre”. The

activity of the respiratory centre is influenced by neural and humoral factors. Among the most important humoral influences are the concentrations of CO_2 and hydrogen ions in the tissue fluids of the respiratory centre, and the rate of firing of receptors in the carotid and aortic bodies (the chemoreceptors). The chemoreceptors are responsive to changes in CO_2 , hydrogen ions and O_2 in the blood. During moderate changes in metabolic rate (as, for example, during aerobic physical exercise), control of ventilation is primarily accomplished by the response of the chemoreceptors and of the respiratory centre to the increase in the CO_2 and hydrogen ion concentration in the blood (Wasserman, Whipp, Casabury, Golden & Beaver, 1979). Although the precise nature of the role of humoral and neural factors in respiratory control remains somewhat obscure, it is a well-established fact that ventilation increases during aerobic exercise to a degree that is directly proportional to the production of CO_2 in the tissues. This increase in ventilation, in turn, results in an increase in the rate of elimination of CO_2 via the lungs. Via this feedback loop, the arterial pCO_2 and pH levels are normally kept within a very narrow range (Gennari & Kassirer, 1982).

Cortical and other influences

Most psychophysiological studies are focused upon behavioural conditions in which there is no, or only a small degree of, physical activity. Although these conditions often involve only minor alterations in the rate of CO_2 production, control of respiration appears, under these circumstances, also to be primarily exerted by humoral factors. However, neural factors may often override metabolic respiratory control mechanisms (von Euler, 1977). Among these are cortical and forebrain mechanisms that act upon the respiratory centre and the respiratory muscles (Bass & Gardner, 1985a; von Euler, 1977). Neural influences are responsible for respiratory changes during the sleep/wake cycle, arousal, cognitive functioning, emotional states and speech (Bass & Gardner, 1985a).

Under most circumstances, the interplay of neural and humoral influences upon respiration seems to be well coordinated. During normal speech, for example, delicate control processes serve to prevent hypo- or hyperventilation. However, under certain emotional and stressful conditions, the coordination of respiratory control mechanisms may break down. This type of breakdown is characterized by exaggerated ventilatory activity (hyperventilation), which causes more CO_2 to be eliminated from the body than is produced by metabolic processes. As a consequence, arterial pCO_2 levels decrease below normal values, leading to a state of respiratory alkalosis and hypocapnia. Largely owing to the alkalosis, hyperventilation may produce a number of adverse physiological and subjective effects. Among the most important effects are reductions in cerebral and myocardial perfusion and O_2

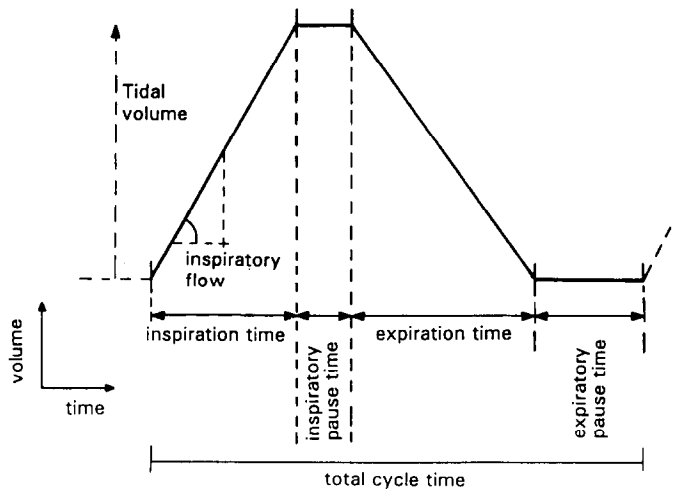


Fig. 1. Time and volume components of the breathing cycle. See text for an explanation.

delivery, ECG and EEG changes, an increase in sympathetic activity, withdrawal of cardiac vagal tone, muscle spasm and tetanus (Grossman & Wientjes, 1989). Subjectively, hyperventilation may elicit numerous somatic and psychological symptoms, such as cardiac and respiratory complaints, dizziness, tingling sensations, in addition to feelings of confusion, anxiety, and panic (Grossman & Wientjes, 1989). Hyperventilation may either be produced voluntarily or may be triggered by stimuli of a threatening, emotional, aversive or painful nature (Grossman & Wientjes, 1989), by passive body movement (Dixon, Steward, Mills, Varvis & Bates, 1961), by immersion in cold water (Cooper, Martin & Riben, 1976), or by a number of pharmacological and pathological conditions (Gennari & Kassirer, 1982).

The breathing cycle

The respiratory cycle consists of an inspiration phase, followed by an expiration phase. There sometimes is a pause between termination of inspiration and onset of expiration, and often between termination of expiration and onset of the next inspiration (Fig. 1). The breathing cycle can be characterized schematically by the following parameters: tidal volume (V_T) (i.e. the volume that is displaced during one breath), duration of inspiration (T_I), duration of expiration (T_E), and total cycle duration (T_{TOT}). Sometimes, the durations of the inspiratory and expiratory pauses (P_I and P_E) are also included in the analyses. Another, more traditional type of analysis only

involves V_T and respiratory rate (RR). However, the limitation of the latter type of analysis is that it does not provide sufficient information concerning the mechanisms which determine V_T and RR . For example, a change in V_T can be accomplished by a change in the rate of inflow of air during inspiration, by a change in T_I , or both. Similarly, RR can be altered by changes in T_I or T_E , or both.

The mechanisms that control the various components of the breathing cycle have, as yet, not all been identified. However, upon the basis of what is currently known, the following picture may be drawn. At least two interacting mechanisms appear to be involved in the regulation of the breathing cycle: a central *inspiratory drive mechanism*, which determines the intensity of the inspiratory stimulus, and a *periodic rhythm generator* which cyclically switches the drive mechanism on and off (Bradley, 1977; von Euler, 1977; Gautier, 1980; Milic-Emili, Grassino & Whitelaw, 1981; Milic-Emili & Grunstein, 1976). Any increase in inspiratory drive manifests itself primarily in an increase in V_T/T_I . The effects of changes in inspiratory drive upon timing seem to be secondary. As long as the inspiratory drive does not change, V_T/T_I remains relatively constant, but there may, at the same time, be marked breath-by-breath variations in V_T and RR (Bradley, 1977; Milic-Emili, Grassino & Whitelaw, 1981). The rhythm generator (which has a variable periodicity) is one of the main determinants of the duration of the inspiratory phase of respiration. However, if V_T increases greatly (as may be the case when V_T/T_I is high), the rhythm generator is overruled, and termination of inspiration is brought about by an afferent vagal reflex, that is triggered by the stretch receptors in the lungs (Clark & von Euler, 1972). Expiration, on the other hand, appears to be under the control of other mechanisms: its duration is primarily a function of the degree of filling of the lungs that was accomplished during the previous inspiration, and of the resistance that the outflow of air encounters (Clark & von Euler, 1972). Therefore, expiration may, under many conditions, be regarded as a passive process.

The contribution of the central inspiratory drive and timing mechanisms to any given breathing cycle may be assessed by the following expression (Gautier, 1980; Milic-Emili, Grassino & Whitelaw, 1981; Milic-Emili & Grunstein, 1976):

$$V_{\text{MIN}} = RR \times V_T = \frac{V_T}{T_I} \times \frac{T_I}{T_{\text{TOT}}}$$

where V_{MIN} is the respiratory minute volume (the total volume of air that is displaced in a 1 min period), the mean inspiratory flow rate (V_T/T_I) is an index of the *intensity* of the central inspiratory drive, and T_I/T_{TOT} is a dimensionless number reflecting the *timing* of respiratory control mechanisms. The latter parameter is often referred to as the inspiratory duty cycle.

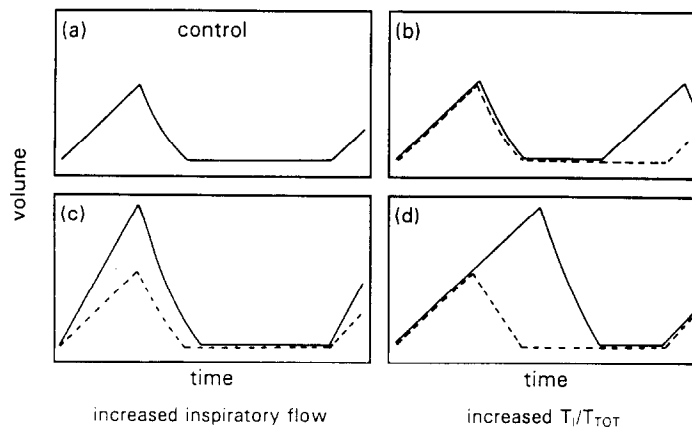


Fig. 2. Schematic spirometry illustrating ways in which ventilation can be increased. (a) Control spirogram. It is reproduced as the broken line in (b) – (c). (b) Inspiratory phase as in (a), but duration of expiration is shortened. (c) Duration of respiratory phases as in (a), but mean inspiratory flow is increased. (d) Mean inspiratory flow and total cycle duration as in (a), but duration of inspiration is increased and duration of expiration is decreased. (Adapted from Milic-Emili, Grassino and Whitelaw, 1981.)

It follows from the equation that a change in ventilation can result from a change in V_T/T_I , in T_I/T_{TOT} , or in both (Milic-Emili, Grassino & Whitelaw (1981); see also Fig. 2). A change in T_I/T_{TOT} occurs when the relationship between T_I and T_E changes. This may occur as a result from a change in T_E (Fig. 2(b)), from a change in T_I , or in both (Fig. 2(d)). Furthermore, ventilation may change as a result of a change in V_T/T_I (Fig. 2(c)). The intricate interplay of these two types of control mechanisms may explain why respiration displays the wide and complex degree of patterning that may be observed among different behavioural conditions, and among different personality types.

Measurement techniques

Time and volume components

Time and volume components of respiration can be measured by a variety of techniques. These range from relatively intrusive techniques such as spirometry and pneumotachography to indirect measurement of respiratory parameters by means of bands or strain gauges. Although they provide very accurate assessment of the timing and volume components of the breathing

cycle, the major disadvantage of intrusive techniques is that they typically employ equipment that adds dead space and resistance to breathing (e.g. mouth piece, facemask or tubes with valves). It has been shown that V_T and RR can be greatly altered during spontaneous breathing with this equipment (Askanazi et al., 1980; Gilbert, Auchincloss, Brodsky & Boden, 1972). On the other hand, techniques employing single strain gauges or thermistors only provide crude and often inaccurate information concerning respiratory volume, and must therefore be considered to be inadequate. Recently, however, unintrusive techniques have become available that enable quantitative assessment of both time and volume components of respiration (Chadha et al., 1982; Cohn et al., 1982; Morel, Forster & Suter, 1983; Sackner, 1980; Watson, 1980). These techniques employ measurement of the separate motions of the rib cage and the abdomen. An essential element of these techniques is the calibration procedure that is used to estimate the volume-to-motion coefficient (VMC) for the rib cage and abdomen signals. The calibration technique is based upon a model that was introduced by Konno & Mead (1967), who argued that the chest wall could be considered as a system which could move with only two degrees of freedom. According to their model, the volume of air that is displaced during respiration can be calculated on the basis of the circumference changes of the rib cage and the abdominal compartments. However, among different individuals, the contribution of both compartments to the total volume displacement may be different. Therefore, both compartments may be considered to have variable gains. These gains (or VMCs) are determined during the calibration procedure. Based upon Konno & Mead's (1967) model, a number of different calibration techniques have been developed (Chadha et al., 1982; Gribbin, 1983; Morel, Forster & Suter, 1983) that employ a calibration session during which the subject either is breathing with a fixed and known volume, or during which V_T is simultaneously measured by means of a pneumotachograph or a spirometer. The rib cage and abdominal VMCs are estimated by means of multiple regression. The general model describing the estimation is (Gribbin, 1983):

$$\text{Volume} = xRC + yAB + e$$

where RC and AB refer to the rib cage and abdominal signals, respectively, and the coefficients x and y are the VMCs for these signals. After proper calibration, measurement of rib cage and abdominal circumference changes provide a reasonably accurate estimation of the time and volume components of respiration (Chadha et al., 1982; Morel, Forster & Suter, 1983), although they are not free from movement artefacts. Slipping of the bands can often be minimized either by taping them to the skin or by employing an elastic vest that is worn over the bands. Partially due to slipping of the bands, however, the validity of the technique diminishes over time: over a measure-

ment period of 5–10 min., the average error of the volume estimate has been found to be about 6%, provided that the posture of the subject remains the same. During longer periods of measurement (2 h), the volume measurement error may increase to 15–20% (Folgering, 1990). Frequent recalibration of the measurement system (e.g. after each experimental condition or measurement session) is therefore strongly recommended. Each of the techniques mentioned has its specific advantages and disadvantages. Inductive plethysmography seems to be less sensitive to movement artefacts than other systems (Morel, Forster & Suter, 1983), but the oscillator is electrically unstable unless it is preheated for a minimum of 30 min. Furthermore, the high-frequency oscillations that are generated by the coils in the bands may interfere with other physiological measurements that operate in the same frequency domain (e.g. impedance cardiography).

Artefacts

During measurement of respiration, two sources of artefacts may be encountered that have to be dealt with: speech and movement. Of course, speech is not an artefact in the proper sense; rather, its influence upon the morphology of the breathing cycle reflects the irregular expiratory movement of air through the glottis that is typical of voice intonation. However, analysis of the respiratory signal during speech in terms of volume and time components is notoriously difficult. Therefore, speech should be avoided as much as possible during the measurements, or the analyses should be restricted to silent epochs. Artefacts caused by movements, which may occur particularly frequently during ambulatory monitoring, may to some degree be dealt with. Anderson & Frank (1990), for example, have employed a digital bandpass filter, that allows frequencies that are common during breathing (6–40 cycles min^{-1}) to pass unhindered, but attenuates higher and lower frequencies, that may be associated with movement.

End-tidal $p\text{CO}_2$

Regarding gas exchange measures, we will only discuss measurement of end-tidal and transcutaneous $p\text{CO}_2$. Measurement of O_2 consumption and CO_2 production is beyond the scope of this paper. For detailed accounts of the techniques regarding these measures, the reader is referred to Langer et al. (1985a).

Direct measurement of arterial $p\text{CO}_2$ involves intra-arterial puncture, which is commonly regarded as too hazardous to be performed in psychophysiological studies. However, in the normal lung, alveolar $p\text{CO}_2$ is considered to be a valid approximation of arterial $p\text{CO}_2$ (Gardner,

Meah & Bass, 1986). Alveolar $p\text{CO}_2$, in turn, can be estimated by measuring the $p\text{CO}_2$ at the end of a normal expiration (end-tidal $p\text{CO}_2$). The most commonly used equipment for measurement of end-tidal $p\text{CO}_2$ is the infrared gas analyser. During the measurements, expired air is sampled through a heated sample tube, which is either attached to an open face mask or to one of the subject's nostrils. The CO_2 content of the sampled air is analysed in a sample cell and the electrical output is converted into one of the conventional physiological units (mmHg, Torr, kPa, percentage, etc.). When the CO_2 values are expressed in partial pressure units, they should be corrected for variations in the barometric pressure. Care should be taken in selecting the proper gas sample flow rate; sample rates that are too high may cause mixing of the expired air with room air (resulting in underestimation of the true end-tidal $p\text{CO}_2$), whereas sample flow rates that are too low may increase the transit time of the air from the face mask to the sample cell beyond acceptable limits, and may even cause a build-up of the CO_2 concentration in the mask cavity (resulting in overestimation of the true end-expiratory $p\text{CO}_2$). Furthermore, the CO_2 waveforms should be carefully inspected before being included in the analyses. When respiration is very shallow, the expired air will consist mainly of gases that have not been in direct contact with the alveolar membrane and, therefore, the end-tidal $p\text{CO}_2$ will, in this case, not be representative of the true alveolar $p\text{CO}_2$. Breaths with a low percentage of alveolar air can easily be recognized by the fact that the CO_2 waveform does not reach a horizontal plateau but, rather, increases steeply and then breaks off suddenly (Bass & Gardner, 1985b). Such breaths should not be included in the analyses. Before use, the infrared gas analyser should be properly calibrated. Calibration of the CO_2 analyser requires two steps. First a sample of CO_2 -free gas is used to set the zero-control to read zero. Then, a sample of gas of a known concentration (5%, for example) is used to set the gain of the system to read that concentration. Often, the 0% calibration gas is replaced by room air, which under normal, well-ventilated conditions contains about 0.03% CO_2 .

Although assessment of end-tidal $p\text{CO}_2$ may appear to be rather complicated and intrusive, this type of measurement has in fact repeatedly been employed in applied settings. For example, Harding (1987) measured end-tidal $p\text{CO}_2$ (and CO_2 production) among pilots during flight. In his study, the sample tube was connected to the oxygen mask of the pilot. The problem of pressure changes during flight was overcome by employing repeated automatic in-flight calibrations.

Transcutaneous $p\text{CO}_2$

Recently, new techniques have become available, employing transcutaneous measurement of arterial $p\text{CO}_2$ by means of a heated electrochemical

sensor (Severinghaus, 1960). The output voltage of the sensor is proportional to the logarithm of the CO_2 concentration in the skin tissues. Heating the sensor to temperatures up to 45°C causes vasodilatation of the capillaries in the skin and arterialization of the capillary blood. Therefore, the CO_2 concentration in the skin tissues is assumed to be representative of the arterial pCO_2 (Severinghaus, Strafford & Bradley, 1978). The validity of the technique has been tested both in vitro and in vivo. Several in vivo studies have shown that the between- and within-subject correlations between arterial and transcutaneous pCO_2 vary between 0.85 and 0.99 (Cherian, Helms, Paky, Marsden & Chiu, 1986; Mindt, Eberhard & Schaefer, 1982; Pilsbury & Hibbert, 1987; Wimberley, Pedersen, Olsson & Siggaard-Andersen, 1985). However, the transcutaneous pCO_2 values are consistently higher than the arterial values, probably because the heating increases the pCO_2 level in the blood and elevates the skin metabolism. Therefore, transcutaneous values are often corrected to obtain estimated arterial values (Cherian et al., 1986). The main problem with this method, however, is caused by its response characteristics. Dependent upon the sensor temperature, the time lag of the response of the sensor to a change in arterial pCO_2 may range between 30 s and 2–5 min (Mindt et al., 1982; Pilsbury & Hibbert, 1987). On the other hand, this method has been shown to be very useful for ambulatory monitoring of pCO_2 (Hibbert, 1986; Pilsbury and Hibbert, 1987). Its reliability and its value for research purposes, however, remain to be assessed precisely.

Analysis techniques

In principle, three types of approaches are available for computer analysis of the respiratory signal: (A) quantitative time domain analysis, (B) qualitative analysis, and (C) spectral analysis. We shall briefly discuss all three approaches.

Quantitative time domain analysis

This technique determines the parameters that quantitatively describe the respiratory cycle (i.e. the time and volume components) by measuring peak amplitudes and the duration of the time components. For this type of signal analysis, relatively straightforward computational techniques can be employed, such as peak detection or assessment of turning points in the digitized respiratory signal (e.g. Wientjes, Grossman & van der Meyden, 1988). This approach has been employed in breath-by-breath analyses of combinations of different physiological signals (e.g. breath-by-breath analysis

of cardiopulmonary or cardiorespiratory measures, see Langer et al., 1985a; Wientjes, Grossman & van der Meyden, 1988).

Qualitative analysis

A second technique, which is currently under development, is designed to evaluate respiratory patterns qualitatively (Boiten, Frijda & Wientjes, 1992). This potentially important technique includes assessment of different types of irregularities in the respiratory pattern (e.g. sighs, periods of apnoea, etc.), and of qualitative characteristics of respiratory behaviour during inspiration and expiration (e.g. concavity and convexity of the inspiratory and expiratory curves). The feasibility and reliability of this type of analysis, however, remain to be determined. Other approaches have employed harmonic analysis (fast Fourier transform) to quantify the shape of the airflow profile (Benchetrit et al., 1989). One of the promising possibilities of this type of analysis is that it may improve our understanding of the finer details of respiratory control to an important degree.

Spectral analysis

A third type of analysis also employs spectral analytic techniques. This approach is not aimed at analysing the frequency characteristics of the respiratory curve, but rather, at determining the periodicity of the respiratory control process. It should be noted, however, that much important information concerning control aspects of respiratory behaviour is "drowned" in spectral analysis. First, spectral analysis typically yields information concerning periodicities in biological steady-states. Under many behavioural conditions and among many individuals, however, respiratory behaviour is highly variable. Assessment of the variability of respiratory parameters is, in fact, an important tool in the study of behavioural and emotional influences upon respiratory control (Grossman & Wientjes, 1989). For this purpose, time domain analysis would clearly seem to be superior to spectral analysis. Second, spectral analysis integrates both amplitude and frequency components of an analogue signal into a single measure: the power density function. As we have seen, however, volume and time components of respiration may often vary independently (Milic-Emili et al., 1981). Therefore, in contrast to time domain analyses, measures derived from spectral analysis of respiratory signals only allow for a limited description of respiratory behaviour. On the other hand, when combined with other spectrally derived measures (such as heart rate or blood pressure variability), the power density function of the respiratory signal permits calculation of coherence and transfer functions. These measures may provide insight into specific aspects of the

covariation of respiration with other physiological processes (see Mulder, 1988).

Respiratory measures in psychophysiological research

Inclusion of respiratory measures in psychophysiological studies has been shown to be relevant for several research areas. These include mental effort and stress, emotional states, personality measures, psychosomatic and clinical affective disorders, and biofeedback. In addition, respiratory measures have been employed in a small number of applied studies, which have investigated the effects of specific demands and stressors. We shall first discuss a number of important studies that have included respiratory measures. Then, we shall try to sketch the perspective that might be offered by a methodologically sound application of respiratory measurement and of advanced analysis approaches. It should be noted, that we shall not attempt to provide an exhaustive review of the literature. Reviews can be found in Bass & Gardner (1985a), Grossman (1983) and Grossman & Wientjes (1989).

Mental effort and stress

A growing number of studies have assessed respiratory responses during the performance of demanding and stressful mental tasks. Although the respiratory parameters that were measured in these studies, and the methods that were employed, vary widely, the results show a remarkably similar pattern. Effortful or stressful mental task performance is generally found to be associated with an increase in RR and V_{MIN} , and with a decrease in V_T and in the variability of respiration (e.g. Allen, Sherwood & Obrist, 1986; Carroll, Turner & Hellawell, 1986; Carroll, Turner & Rogers, 1987; Kagan & Rosman, 1964; Langer et al., 1985b; Sims, Carroll, Turner & Hewitt, 1988; Svebak, 1982; Turner & Carroll, 1985; Turner, Carroll & Courtney, 1983; Wientjes, Grossman, Gaillard & Defares, 1986). [Details of the respiratory response to demanding mental tasks can best be illustrated by a recent study that employed extensive component analysis of respiration (Wientjes et al., 1986).] In this study, 44 male subjects performed a memory comparison task under three differentially demanding conditions: (A) no feedback (NFB), (B) trial-by-trial feedback of performance (FB), and (C) an "all-or-nothing" condition (AON), where feedback was provided in a similar manner as in the FB condition, but where the subjects had to improve their best previous performance level in order to receive a monetary reward. The NFB and FB conditions were presented in counterbalanced order, and the AON condition always came last. Respiration was measured via inductive plethysmography, and respiratory parameters included V_T , T_I , T_E and T_{TOT} .

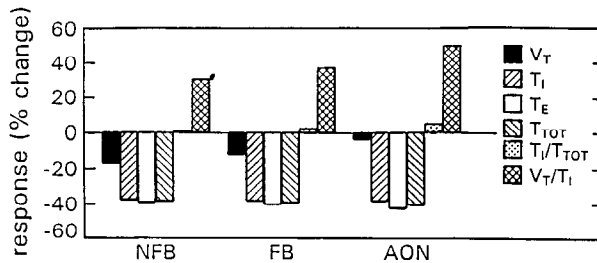


Fig. 3. Response patterns of respiratory parameters during three conditions of a memory-comparison reaction time task (percentage change from baseline). NFB = no feedback; FB = feedback; AON = "all or nothing". See text for explanation.

On the basis of these measures, mean inspiratory flow rate (V_T/T_I) and duty cycle time (T_I/T_{TOT}) were determined. A summary of the results of the study can be found in Fig. 3.

In terms of the model of respiratory control mechanisms of Milic-Emili and his coworkers (1976, 1981) that was discussed earlier, the results of the Wientjes et al. (1986) study suggest that the increase in respiratory activity that was evoked during the task performance was a consequence of (A) a greatly enhanced central inspiratory drive and (B) a small shift in the periodicity of the central timing mechanism, causing a greater portion of the respiratory cycle to be occupied by inspiration. The magnitude of both effects increased as the task demands increased. As can be seen in Fig. 3, the task-related increase in V_T/T_I was brought about by an interaction between the responses of V_T and of T_I : whereas the degree of shortening of T_I remained more or less constant across conditions, the decrease in V_T became less pronounced as the task demands increased, and V_T reached, during the most demanding condition (AON), a level that was nearly equal to the initial resting value. Interestingly, the results of this study imply that the most commonly used respiratory parameter (RR), was unresponsive to differences in the task demands. In summary, this study suggests that the respiratory pattern during demanding mental task performance is characterized by relatively fast, shallow breathing, with a high mean inspiratory flow rate. As demands increase, the breathing pattern typically becomes less shallow, and mean inspiratory flow rate is even further augmented, but there is no change in the rate of breathing. It may also be concluded, on the basis of these results, that the respiratory response to psychological stress can only be adequately assessed if the proper respiratory parameters are measured.

Emotions and affect

Although there is a long standing research tradition with respect to links between emotions and respiratory responses, the available studies generally lack methodological sophistication. Therefore, our knowledge concerning the influence of emotions upon respiration is generally restricted to traditional respiratory parameters such as *RR* and (often unreliable estimates of) V_T . Some recent studies have also included measurement of end-tidal pCO_2 . Efforts to differentiate between respiratory patterns during different emotional states have, probably partly as a consequence of inadequate methodology, and partly as a result of conceptual confusion about emotion categories, produced somewhat conflicting results. In spite of its obvious limitations, this extensive area of research has nevertheless yielded many suggestions concerning the behavioural significance of respiratory responses. On this basis, we shall attempt to draw a few broad lines linking emotional states and respiratory patterns.

The first crude distinction that can be made is between respiratory hyper- and hypofunction. Along this response dimension, the respiratory pattern seems primarily to vary along a continuum that ranges from excited and action-oriented states, via attentive, alert states, to inhibited or low arousal states (Boiten, Frijda & Wientjes, 1992; Dudley, Holmes, Martin & Ripley, 1964). Another crude distinction (that is not independent of the first) is between metabolically appropriate ventilation (normoventilation) and inappropriate ventilation (hyperventilation). Whereas normoventilatory responses seems to be characteristic of effortful adaptation to psychosocial challenges (i.e. active coping) (Allen et al., 1986; Langer et al., 1985b; Wientjes et al., 1986), hyperventilation has been shown to occur during situations where very few, if any, active coping possibilities exist, such as states of pain or aversive stimulation, apprehension, anxiety and fear, threat and anger (Allen et al., 1986; Dudley et al., 1964; Freeman, Conway & Nixon, 1986; Suess, Alexander, Smith, Sweeny & Marion, 1980). Thus, hyperventilation may probably be regarded as a typical "passive coping" response (Grossman & Wientjes, 1989; Obrist, 1981).

Personality traits, psychosomatic disorders and clinical groups

Several studies have shown that scores on questionnaires measuring neuroticism, trait anxiety, introversion, depression and tendencies to experience psychosomatic complaints are associated with increased respiratory activity, with depressed levels of end-tidal pCO_2 , or with enhanced ventilatory reactivity to stressful situations (e.g. Damas-Mora, Grant, Kenyon, Patel & Jenner, 1976; Damas-Mora, Souster & Jenner, 1982; Grossman, de Swart & Defares, 1985; McCollum, Burch & Roessler, 1969; Morgan, 1983;

Oken et al., 1962; Shershaw, King & Robinson, 1973; Skarbak, 1970; Wientjes, Grossman & Defares, 1984). Likewise, a number of studies have reported respiratory hyperactivity, irregular respiration and depressed levels of end-tidal $p\text{CO}_2$ among groups of subjects that suffer from hyperventilation syndrome, panic disorder, and related psychosomatic disorders (e.g. Folgering & Colla, 1978; Freeman et al., 1986; Gardner et al., 1986; Grossman & Wientjes, 1989; Wientjes & Grossman, 1992).

There are repeated reports in the literature which suggest that increased levels of psychosomatic symptom reporting may be associated with excessive breathing and hyperventilation (Gardner et al., 1986; Huey & West, 1983; Wientjes & Grossman, 1992; Wientjes et al., 1984; Wientjes, Grossman & de Wolf-Hopman, 1987). As we mentioned earlier, there is, indeed, ample evidence that hyperventilation, hypocapnia and their physiological concomitants can elicit a variety of somatic and psychological symptoms (Clark & Helmsley, 1982; Svebak & Grossman, 1985). However, that does not imply that hyperventilation should necessarily be considered to be the cause these symptoms. There is also firm evidence for the involvement of psychological mechanisms in the formation of symptoms. Several investigations have found systematic relationships between degree of psychosomatic symptom reporting and personality traits such as neuroticism, anxiety and depression (Watson & Pennebaker, 1989; Wientjes, Grossman & Defares, 1984; Wientjes, Grossman & de Wolf-Hopman, 1987). To explain these findings, it has been proposed that anxious and neurotic individuals are more likely to focus their attention on internal, somatic stimuli, rather than on external events. Additionally, these individuals may have a higher tendency towards negative appraisal and negative evaluation of their experiences (Watson & Pennebaker, 1989). As a consequence, they might be particularly sensitized to perceive the bodily sensations that often accompany states of high physiological activation, and to evaluate these sensations in terms of anxiety-provoking symptoms (Grossman & Wientjes, 1989; Watson & Pennebaker, 1989). Thus, investigation of covariations between breathing, symptom reporting, and psychological factors may be very helpful in efforts to disentangle this multiple causation (see Wientjes & Grossman, 1992). Such research could greatly contribute to the understanding, and the prevention, of otherwise unexplained psychosomatic symptoms in daily life and in working environments.

The results of studies on subjects with clinical diagnoses are generally in support of the findings concerning associations between personality traits and respiration. Breathing patterns were shown to differentiate between diagnostic groups such as patients with anxiety neurosis, phobias, depression, psychoses and normals (e.g. Damas-Mora et al., 1976; Damas-Mora et al., 1982; Skarbak, 1970), and have served as a diagnostic tool for assessing such groups.

Biofeedback

Some investigations have suggested that modifying and regulating respiration may help to reduce subjective distress (e.g. Clark, Salkovskis & Chalkley, 1985; Grossman et al., 1985). For example, one study investigated the effect of respiratory biofeedback versus a placebo treatment among groups of hyperventilation syndrome patients (Grossman et al., 1985). Only the biofeedback group showed improvement in respiration (resting *RR* and resting end-tidal $p\text{CO}_2$) and in questionnaire measures (neuroticism, anxiety and psychosomatic symptom score). Thus, modification of the breathing pattern in the direction of slow, normocapnic respiration may, at least to some degree, serve to alleviate distressful feelings and symptoms, and may even be effective in modifying certain aspects of personality measures.

Applied studies

Respiratory measures have only been included in very few applied studies. However, these measures may for several reasons prove to be an important research tool: (A) respiratory responses could help to evaluate the effects of mental workload and of situational stressors, (B) certain respiratory parameters might be used to obtain an approximation of energy expenditure, and (C) assessment of the occurrence of hyperventilation may highlight undesirable and potentially dangerous aspects of the task environment. The few applied studies that have employed respiratory measures have yielded potentially important results. For example, in an extensive recent study by Harding (1987), respiration and end-tidal $p\text{CO}_2$ were measured among pilots during different flight profiles in a high-performance aircraft. The results show that *RR*, peak inspiratory flow and V_{MIN} were increased during all phases of routine flight, and that marked elevations of respiratory activity occurred during manoeuvring flight phases. Mild, but sustained hyperventilation occurred during demanding manoeuvring phases of the flights. Hyperventilation was most pronounced during flight phases that involved high *G* manoeuvres; this was probably partly due to the active methods that were applied by the pilots to increase their tolerance to +*G* acceleration ("straining"). However, the end-tidal $p\text{CO}_2$ values were, generally, well above the range that may be considered dangerous. It should be noted, however, that the research flights did not include stressful phases. It has frequently been suggested that stress during flight might lead to substantial falls in $p\text{CO}_2$, with potentially dangerous consequences (Gibson, 1979; 1984). One of the possible detrimental consequences of hyperventilation may be its pronounced effect upon psychomotor performance (Gibson, 1978), which may be due either to central factors (cerebral hypoxia) or to peripheral factors (tremors

and tetanus). However, there is very little evidence to support Gibson's (1979; 1984) suggestions about the dangers of hyperventilation during flight.

Another series of applied studies has measured end-tidal $p\text{CO}_2$ during exposure to vestibular stimulation (Bles, Boer, Keuning, Vermeij & Wientjes, 1988; Bles & Wientjes, 1988). In one study, which was performed (under a variety of sea states) on board a mine-hunter of the Dutch Navy, there were high within-individual correlations between the degree of motion sickness, state anxiety, and end-tidal $p\text{CO}_2$ (Bles et al., 1988). In contrast to Harding's (1987) findings, the hyperventilation was, among the most sick subjects (who were all experienced sailors), quite substantial. These findings were later confirmed in a laboratory study, although the degree of hyperventilation was much milder there (Bles & Wientjes, 1988). These results suggest that hyperventilation may indeed, as is suggested by the literature on emotional stress, serve as a useful index of the impact of an aversive stressor upon the individual.

Respiratory measures in a behavioural perspective

As we have seen, there is compelling evidence that respiratory responses may, on the one hand, be closely tied to various situational demands, and on the other, to certain characteristics of the individual (e.g. coping style or personality). Given this combined influence of situational- and individual-specific factors, which behavioural framework are we to employ in the interpretation of respiratory response pattern? A first approach might be to find links between appraisal processes, and respiratory responses. Appraisals encompass an individual's estimate about the potential relevance, benignity or positiveness, and stressfulness of the stimulus; furthermore, appraisals involve judgements about the likelihood that the stimulus may have consequences in terms of harm or loss, threat, or challenge (Lazarus & Folkman, 1984). Thereby, appraisals determine, on the one hand, the affective response. On the other hand, appraisal patterns may, on a physiological level, correspond to particular response sets (Frijda, 1986), or functional categories such as "flight or fight" (Cannon, 1929), "intake or rejection" (Lacey and Lacey, 1970), and "passive or active coping" (Obrist, 1981). Response sets or functional categories may, in turn, be specified in terms of the patterning of arousal, activation and inhibition processes (Pribram, 1981). Thus, respiratory responses might be linked to response sets reflecting, for example the (perceived) need for mobilization of effort, need for control, or expectations about loss of control, orientation towards action or inhibition, and approach or avoidance. Along the lines of this type of analysis, it might be possible to develop a framework of behavioural energetics that could be helpful in mapping out links between different environmental conditions and psycho-

logical categories on the one hand, and identifiable respiratory response patterns on the other (Hockey, Coles & Gaillard, 1986).

Dimensions of respiratory responses

Taken together, the results of the studies that were discussed suggest that respiratory responses may vary in at least two response dimensions: (A) with regard to the morphology of the breathing cycle (i.e. pattern and intensity), and (B) in terms of normo-versus hyperventilation (as reflected by estimates of arterial $p\text{CO}_2$ levels). It is precisely this multidimensional nature of the respiratory response, and the long standing failure of psychophysiological tradition to develop or adopt methods that can adequately deal with its complexity, that may be held responsible for the relative lack of progress in respiratory psychophysiology since its promising start at the beginning of this century (e.g. Féléky, 1914; Rehwolt, 1911). However, this multidimensionality of the respiratory response provides, at the same time, unique opportunities for advanced techniques of pattern and intensity analysis to be employed (Stemmler & Fahrenberg, 1989).

The morphology of the breathing cycle

Variation in the morphology of the breathing cycle is operationalized in terms of the patterning of the changes in the time and volume components, or, if the model of Milic-Emili and his coworkers (1976; 1981) is applied, in terms of the drive and timing aspects of respiration. In principle, this response dimension encompasses a wide range of patterns. However, because only very few studies have applied this type of analytic approach, we are still lacking basic knowledge concerning the types of patterns that may be observed under different behavioural conditions, and about the potential merits of this approach. Given these obvious limitations, the available literature may nevertheless be summarized in a manner that is suggestive with regard to the types of correlations that might be found between psychological dimensions and respiratory patterns. It would seem that at least four different respiratory patterns can be distinguished, that may each be linked to specific psychological or behavioural dimensions:

(A) Rapid shallow breathing with a high inspiratory flow rate is often associated with effortful mental task involvement (Wientjes et al., 1986) or with sustained attention (Kagan & Rosman, 1964). This pattern may also be associated with “tense affects”, with “tense expectation” (Boiten, Frijda & Wientjes, 1992), with moderate anxiety or fear (Ax, 1953; Schachter, 1957), and it may be observed among anxious, fearful and tense individuals (Grossman, 1983; Wientjes et al., 1986). Taking all the available evidence into account, rapid shallow breathing thus appears to be correlated with (tense)

anticipation involving attention and alertness, or with consummatory behaviour characterized by some degree of control over effortful action tendencies. Emotionally, this pattern may be associated with a range of affects varying from tension to anxiety. It could be that this manner of respiratory responding is typical for attempts to tune in to behavioural demands that might necessitate a restrained, precise, and goal-directed type of overt physical action.

(B) Rapid, deep breathing with a high inspiratory flow rate, on the other hand, appears to lack this control aspect, and seems typical for exited and aroused states, which are characterized by unrestrained or massive action orientation (Dudley et al., 1964). In this respect, it is noteworthy that rapid deep breathing is also observed during hypnotic suggestion of exercise (Morgan, 1985). This pattern may be associated with a wide range of emotional states, as can be encountered in extreme fear or anger, or in joy and ecstasy. The action orientation aspect may also be aimless or undirected, as, for example, during "mere excitement" (Frijda, 1986). If inhibition of strong action tendencies ("fight or flight") is involved, this type of responding may also encompass the hyperventilatory "passive coping" response (see below).

(C) Relatively little is known about the behavioural and emotional correlations of slow, shallow breathing. However, there are indications that this pattern may be associated with "passive grief" and "depressed affect" on the one hand, or with "calm happiness" and "satisfaction" on the other (Averill, 1969). Thus, this pattern of breathing appears to be primarily indicative of states that are characterized by withdrawal from the environment.

(D) Deep, slow breathing is characteristic of resting states; this pattern may be enhanced during slow-wave sleep (Snyder & Scott, 1972), and during deep relaxation (Grossman, 1983). There are very few observations that have linked this pattern of breathing to other behavioural, or emotional, responses.

Normo- versus hyperventilation

The second response domain in which respiratory behaviour may vary is operationalized in terms of the degree to which the response is in accordance with metabolic activity. The notion of metabolically exaggerated physiological reactivity is quite common in cardiovascular research (e.g. Obrist, 1981), and it has repeatedly been suggested that this type of reactivity may be associated with cardiovascular disease processes (e.g., Krantz & Manuck, 1984). However, regarding metabolically exaggerated respiratory responses, such claims may be encountered less frequently (although Grossman & Wientjes (1985) have linked this type of response to certain functional cardiac disorders). Whether hyperventilation is really harmful to the organism or not, this response certainly signals a profound disruption of normal homeostatic

respiratory functioning. As such, it provides impressive insight into the effective impact which certain psychosocial and other stressors have upon the individual (e.g. Bles & Wientjes, 1988; Freeman et al., 1986; Suess, Alexander, Smith, Sweeny & Marion, 1980). As we have noted earlier, hyperventilatory respiratory activity appears to be strongly associated with an unsuccessful outcome of the coping process, either because the individual fails to mobilize the adequate resources, or because the challenge is too overwhelming. This type of coping is often called "passive coping" because the individual has no other choice than to try to accommodate to the stressor, or to try to reduce its impact as much as possible (Obrist, 1981). Passive coping may often involve strong (but inhibited) action-oriented, or aggressive tendencies. In this respect, it has often been suggested that hyperventilatory responses may functionally be interpreted in terms of preparation for fight or flight (Cannon, 1929; Suess et al., 1980; Dudley et al., 1964). Interestingly, as we have seen, hyperventilation appears not only to be elicited by aversive stressors, but its physiological effects also often seem to be the cause of distressful subjective experiences such as anxiety-provoking psychosomatic symptoms (Gardner et al., 1986; Huey & West, 1983; Wientjes et al., 1984; Wientjes et al., 1987). As has often been pointed out (e.g. Clark et al., 1985), these distressing subjective responses may, via a feedback loop, serve to increase the degree of stress and discomfort that is experienced by the individual, thereby maybe reinforcing the inappropriate physiological response.

Conclusion

As we have seen, reliable and unobtrusive measurement of respiration is not only practically feasible, but respiratory measures may also contribute significantly to psychophysiological research issues. In this respect, one of the most important features of respiratory responses is their multidimensional nature. Given the appropriate techniques for the analysis of respiration in terms of volume and time components, it seems possible to find consistent associations between variations in the patterning and in the intensity of respiratory responses on the one hand, and a range of behavioural conditions on the other. Respiration, however, appears not only to be correlated in an orderly fashion with situational influences, it also covaries in a meaningful way with dispositional psychological factors. Accordingly, respiratory measures might provide unique possibilities to assess individual differences in coping patterns. The most striking feature of respiration, however, might be its close connection to subjective experience. The degree to which breathing is tied to various emotion and affect dimensions, to moods and to a range of psychosomatic symptoms, appears to be unique in psychophysiology. Thus, it

seems that respiratory measures would be a splendid choice in a type of study that has often been advocated (e.g. Hockey et al., 1986), but has only very infrequently been performed: studies that investigate the covariation of physiological and subjective processes.

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